

# Product Backlog - Sprint 1

---

## Project Backlog

This document outlines the product backlog items organized by epics, features, user stories, and priority levels.

## #. Epic

### Features

| User Story   | Priority |
|--------------|----------|
| User Stories | Priority |

## 1. User Login & Sign-in

### Welcome Screen

| User Story                                         | Priority |
|----------------------------------------------------|----------|
| Display Welcome Screen on App Launch for new Users | High     |
| Display system logo and chatbot introduction       | High     |

### User Registration

| User Story                                          | Priority |
|-----------------------------------------------------|----------|
| Provide User Registration Screen                    | High     |
| Allow Users to Register using Email and Password    | High     |
| Allow Users to Register using Google Authentication | High     |
| Store User Details Securely in Database             | High     |

### User Login

| User Story                                        | Priority |
|---------------------------------------------------|----------|
| Provide Secure Login Screen                       | High     |
| Allow Users to Log in using Email and Password    | High     |
| Allow Users to Log in using Google Authentication | High     |
| Validate User Credentials before Login            | High     |

### User Profile Management

| User Story                                 | Priority |
|--------------------------------------------|----------|
| Allow Users to View Profile Information    | High     |
| Allow Users to Edit Profile Information    | High     |
| Store User Preferences for Personalization | High     |

## 2. Chatbot Interface

### Chat Interface

| User Story                                         | Priority |
|----------------------------------------------------|----------|
| Provide Chat Interface for User Interaction        | High     |
| Allow Users to Send Messages to Chatbot            | High     |
| Display Chatbot Responses in Real-Time             | High     |
| Maintain conversation flow between user and system | High     |

### Conversational Understanding using LLM

| User Story                                            | Priority |
|-------------------------------------------------------|----------|
| Process user messages using Google Vertex AI (Gemini) | High     |
| Extract intent and entities from user messages        | High     |
| Maintain dialogue context across multiple turns       | High     |

### Message History

| User Story                                | Priority |
|-------------------------------------------|----------|
| Display chat message history              | High     |
| Allow users to view past conversations    | High     |
| Provide conversation search functionality | High     |

## 3. Agentic AI Orchestration System

### Orchestration Agent

| User Story                        | Priority |
|-----------------------------------|----------|
| Route user intents to appropriate | High     |

|                                                        |      |
|--------------------------------------------------------|------|
| specialized agents                                     |      |
| Maintain conversation context across agent delegations | High |
| Coordinate multi-step workflows                        | High |

### Context Management

| User Story                             | Priority |
|----------------------------------------|----------|
| Store conversation context in memory   | High     |
| Pass context between agents            | High     |
| Maintain user state throughout session | High     |

### Agent Coordination

| User Story                                         | Priority |
|----------------------------------------------------|----------|
| Define agent capabilities and tools                | High     |
| Implement tool calling for agent actions           | High     |
| Handle agent responses and integrate into dialogue | High     |

## 4. Restaurant Discovery Agent

### Restaurant Search

| User Story                                                          | Priority |
|---------------------------------------------------------------------|----------|
| Allow users to search for restaurants by name                       | High     |
| Support multi-criteria filtering (cuisine, price, location, rating) | High     |
| Provide real-time search suggestions                                | High     |

### Restaurant Information Display

| User Story                                               | Priority |
|----------------------------------------------------------|----------|
| Display restaurant details (name, address, phone, hours) | High     |
| Show menu and dietary information                        | High     |
| Display customer reviews and ratings                     | High     |

### Availability Checking

| User Story                         | Priority |
|------------------------------------|----------|
| Check real-time table availability | High     |

|                                             |      |
|---------------------------------------------|------|
| Display time slots with availability status | High |
| Integrate with Firebase for live data       | High |

## 5. Personalized Recommendation Agent

### Knowledge Graph Management

| User Story                                            | Priority |
|-------------------------------------------------------|----------|
| Build Neo4j knowledge graph for users and restaurants | High     |
| Model user-restaurant-cuisine relationships           | High     |
| Store preference weights and affinities               | High     |

### Preference-Based Recommendation

| User Story                                                     | Priority |
|----------------------------------------------------------------|----------|
| Generate recommendations based on user history                 | High     |
| Provide personalized restaurant suggestions                    | High     |
| Explain recommendations contextually in dialogue               | High     |
| Handle cold-start problem through conversational clarification | High     |

### Adaptive Learning

| User Story                                         | Priority |
|----------------------------------------------------|----------|
| Update preference model based on user interactions | High     |
| Refine recommendations with each booking           | High     |
| Track cuisine preferences and dining occasions     | High     |

## 6. Reservation Management Agent

### Availability Checking

| User Story                                   | Priority |
|----------------------------------------------|----------|
| Query restaurant availability from Firebase  | High     |
| Check table status by date and time          | High     |
| Display available time slots and party sizes | High     |

## Reservation Creation

| User Story                                           | Priority |
|------------------------------------------------------|----------|
| Allow users to book tables through chat              | High     |
| Capture party size, date, time, and special requests | High     |
| Confirm reservation with user                        | High     |

## Reservation Management

| User Story                                  | Priority |
|---------------------------------------------|----------|
| Display user's upcoming reservations        | High     |
| Allow users to modify reservation date/time | High     |
| Allow users to cancel reservations          | High     |
| Send reservation reminders                  | High     |

## 7. Payment Handling Agent

### Payment Processing

| User Story                                                  | Priority |
|-------------------------------------------------------------|----------|
| Provide secure payment gateway integration (Stripe/PayHere) | High     |
| Process credit/debit card payments                          | High     |
| Support local payment methods (PayHere, Dialog eCash)       | High     |

### Payment Confirmation

| User Story                          | Priority |
|-------------------------------------|----------|
| Display payment receipt             | High     |
| Send payment confirmation email/SMS | High     |
| Store payment records securely      | High     |

## 8. Firebase Real-Time Data Management

### Restaurant Data Management

| User Story                               | Priority |
|------------------------------------------|----------|
| Store restaurant profiles in Firestore   | Medium   |
| Manage restaurant menu data in real-time | Medium   |
| Update restaurant availability status    | Medium   |

## Reservation Data Management

| User Story                             | Priority |
|----------------------------------------|----------|
| Store user reservations in Firestore   | Medium   |
| Sync reservation status across devices | Medium   |
| Archive completed reservations         | Medium   |

## User Data Synchronization

| User Story                                          | Priority |
|-----------------------------------------------------|----------|
| Sync user profile data in real-time                 | Medium   |
| Store user preferences and interaction history      | Medium   |
| Maintain user-restaurant interaction logs for Neo4j | Medium   |

## 9. Restaurant Provider Management System

### Restaurant Profile Management

| User Story                                        | Priority |
|---------------------------------------------------|----------|
| Allow restaurants to create/edit profiles         | Medium   |
| Manage restaurant information and contact details | Medium   |
| Upload restaurant photos and branding             | Medium   |

### Menu Management

| User Story                                         | Priority |
|----------------------------------------------------|----------|
| Allow restaurants to create/edit menu items        | Medium   |
| Upload item descriptions, prices, and dietary info | Medium   |
| Manage item availability and special offers        | Medium   |

### Table Availability Management

| User Story                              | Priority |
|-----------------------------------------|----------|
| Set table configuration and capacities  | Medium   |
| Manage table availability schedules     | Medium   |
| Handle table reservations and occupancy | Medium   |

## 10. Admin Management System

### User Management

| User Story                   | Priority |
|------------------------------|----------|
| View all registered users    | Medium   |
| Monitor user activity        | Medium   |
| Handle user support requests | Medium   |

### Restaurant Management

| User Story                                      | Priority |
|-------------------------------------------------|----------|
| Verify and approve new restaurant registrations | Medium   |
| Monitor restaurant profile data                 | Medium   |
| Handle restaurant disputes and support          | Medium   |

### System Monitoring

| User Story                           | Priority |
|--------------------------------------|----------|
| Monitor agent performance metrics    | Medium   |
| Track system uptime and availability | Medium   |
| View error logs and system health    | Medium   |

## 11. Security and Access Control

### Authentication and Authorization

| User Story                                       | Priority |
|--------------------------------------------------|----------|
| Implement OAuth 2.0 for Google authentication    | Medium   |
| Implement email/password authentication          | Medium   |
| Implement OTP verification for sensitive actions | Medium   |
| Implement role-based access control (RBAC)       | Medium   |

### Data Security

| User Story                                     | Priority |
|------------------------------------------------|----------|
| Encrypt sensitive data in transit and at rest  | Medium   |
| Implement secure password hashing              | Medium   |
| Manage API keys and secrets securely           | Medium   |
| Implement data privacy compliance (GDPR/local) | Medium   |

## 12. Notification System

### Reservation Notifications

| User Story                                     | Priority |
|------------------------------------------------|----------|
| Send reservation confirmation SMS/Email        | Medium   |
| Send reminder notifications before reservation | Medium   |
| Notify users of reservation changes            | Medium   |
| Send post-dining follow-up messages            | Medium   |

### System Notifications

| User Story                                   | Priority |
|----------------------------------------------|----------|
| Push notifications for app events            | Medium   |
| In-app notification center                   | Medium   |
| Alert users of system updates or maintenance | Medium   |

## 13. System Integration and Deployment

### Backend Integration

| User Story                                    | Priority |
|-----------------------------------------------|----------|
| Integrate Node.js API with FastAPI AI service | Medium   |
| Integrate Firebase with all backend services  | Medium   |
| Integrate Neo4j with AI service               | Medium   |
| Implement service-to-service authentication   | Medium   |

### System Deployment

| User Story                        | Priority |
|-----------------------------------|----------|
| Containerize services with Docker | Medium   |
| Deploy to Google Cloud Platform   | Medium   |
| Set up CI/CD pipeline             | Medium   |
| Configure monitoring and logging  | Medium   |